

Raghav Senthil Kumar

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EDUCATION

University of Illinois Urbana-Champaign

Champaign, IL

B.S. in Computer Science and Statistics

Expected: May 2029

- **Coursework:** Computer Science I (Java), Computer Science II (C++), Data Science & ML (Python), Calculus III
- **Activities:** Disruption Lab, SIGRobotics, AI Alignment, Agentic AI, BuildIllinois, Innovation LLC

EXPERIENCE

Founding Software Engineer | Clusion

September 2025 – January 2026

- Supported an *Agile* team of **10 engineers** building cloud-hosted document search on *AWS* for **10K+ site visitors**
- Flagged **30%** of evidence documents as identical with a de-duplication algorithm, supporting overall data quality
- Analyzed the word-count distribution of tag, citation and evidence text to filter statistical outliers with **IQR**

Product Development Intern | Jetson

January 2025 – February 2025

- Generated insights from **7 iterations** of UI/UX testing across **3 experimental features** (**100K+ users**)
- Recommended data-driven strategies to optimize user engagement in a **5-page report** delivered to Jetson's CEO
- Supported decision-making with **A/B testing** on 30 test users, cutting *Figma* onboarding from **12 to 5 pages**

Data Analytics Intern | STEM E

June 2024 – July 2024

- Analyzed **3 years** of TikTok data (**200+ posts**) to identify trends using *Python (pandas)*, *Power BI* and *Excel*
- Found out that science “fun facts” posts made up just **7%** of all posted content yet earned **69.3%** of all views
- Translated the analysis into insights and data-driven recommendations, pitching a “fun facts” strategy to the Executive Director, who adopted it; engagement rose **24%** (8K+ views), supporting business decision-making

PROJECTS

Truck Pricing VLM | [Live Website](#) | [Source Code](#)

September 2026

- Led a team of **4** to **2nd at UIUC Founder's Hackathon** with a pricing algorithm for **Kamion (YC S22)**
- Cleaned and prepared data from **15,000** scraped truck-listing images via de-duplication and non-USD filtering
- Engineered a generative AI agent (*Claude Sonnet 4.6*) with a *JSON schema* to extract truck details from photos
- Tested and refined the extraction schema until **10 consecutive images** were correct on each of **3 features**
- Fit a *quantile regression* model on the extracted features to predict prices at **87% accuracy** across **30 samples**

SchoolPulse AI | [Live Website](#) | [Source Code](#)

June 2026

- Developed **3 edge AI tools** that automate the collection of a school's energy, water and food consumption data
- Built a *Python FastAPI* backend that ingests live water, waste and energy data from all **3 tools** for analysis
- Applied **4 Random Forest** machine learning models on the ingested live data from all **3 tools** to detect anomalies and deliver actionable steps via a *Gemma Voice LLM* on a *Next.js* dashboard, turning live data into insights
- Directed a team of **5** from initial ideation to the final presentation as a **USAII Global Hackathon Finalist**

Citation Chain | [Live Website](#) | [Source Code](#)

June 2026

- Queried a *PostgreSQL* database hosted on *Supabase* with *SQL* to analyze papers by field, year and citation count
- Rendered each paper's outbound citations as an interactive *React Flow* graph spanning **26 research fields**
- Deployed a **RAG** AI agent (*Claude Sonnet 4.5*) that uses vector search to retrieve session-scoped documents
- Pulled papers from *OpenAlex* and filled missing metadata from *Semantic Scholar* in a two-API data pipeline

EcoBin | [Research Paper](#) | [Source Code](#)

March 2026 – May 2026

- Designed, trained, tested and evaluated a two-stage *EfficientNetV2-S* deep learning neural network (**4.39M** parameters) in *PyTorch* on **15K+ images** to classify waste items into 4 disposal pathways at **96.13% accuracy**
- Prepared data for machine learning with a synthetic pipeline (*U2-Net*, alpha-compositing) generating **9K** images
- Shared the work as an *arXiv* paper and **2 Kaggle datasets**; **placed 2nd at Arizona State Science Fair**

SKILLS

Programming Languages: Python, SQL, Java, JavaScript, C++, HTML/CSS

Frameworks/Platforms: PostgreSQL, Supabase, MySQL, AWS, Google Cloud, Elasticsearch, FastAPI, Next.js

AI/ML: Claude API, scikit-learn, PyTorch, Gemma 4, Hugging Face, TensorFlow, Keras, Supernanny

Data & Analytics: Power BI, Microsoft Excel, pandas, NumPy, A/B Testing, Hootsuite

Concepts/Practices: Data Analysis, Data Insights, Data-Driven Decision-Making, Data Visualization